

ISFANDYAR QURESHI

IOT / EMBEDDED ENGINEER

PROFILE

Experienced Embedded Software Engineer adept in firmware development and PCB design for IoT. Proficient in C/C++ and Python, with expertise in STM32 microcontrollers and FreeRTOS. Skilled in CI/CD pipeline setup for Raspberry Pi and ESP32. Developed successful energy monitoring devices utilizing ESP32 and Raspberry Pi. Strong background in digital IoT hardware design

PROFESSIONAL EXPERIENCE

IOT ENGINEER

MAVARICK ,IRELAND

June 2023 - April 2024

- Extensively worked on **PyODBC** and **FreeTDS** driver for MySQL
- Resolving memory leaks and database connection issues
- Designed hardware with **ESP32** and GSM module 4-layer **PCB**.
- Implemented **CI/CD** pipeline and over-the-air updates using Tailscale
- Made ready to install OS for raspberry pi for hassle free IOT device installation
- Implemented **HTTPS** and **WiFi** provisioning, integrating **AWS IoT** on **ESP32** using **ESP-IDF**.

Embedded ENGINEER

Rapidev ,Pakistan

Sep 2022 - June 2023

- Lead team of engineers to design a software define radio which
- involved **stm32 ,GPS** and transceiver chip
- Design of networking protocol on top of 802.15.4 IEEE stack
- Design and implementation of **software architecture**
- Hardware in Loop testing of product

Embedded ENGINEER

Teresol Pvt LTD, Pakistan

Jan 2019-Dec 2021

- Routing of 6 layer **PCB** and making custom footprints for hardware parts
- Writing of **C firmware** which controlled boot sequence of board and Read data from dual port ram
- Initial board inspection and test firmware writing and **hardware in loop testing**

CONTACT

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EDUCATION

Air University 2013-2017 Pakistan

Bachelor's degree Electrical and Electronics Engineering

Air University 2013-2017 Pakistan

Specialization in electronics

SKILLS/EXPERTISE

- **ALTium**
- **AWS IoT Core**
- **Bash scripting**
- **C/C++**
- **CMAKE**
- **CUBE-IDE**
- **ESP-IDF**
- **ESP32**
- **Firmware development**
- **Free-RTOS**
- **GitHub**
- **GRAFANA DASHBOARD**
- **JIRA**
- **KICAD**
- **PCB hardware design**
- **PYTHON**
- **STM32**
- **Software architecture**
- **Unit testing Unity/CppuTest/Pytest**
- **VSCODE**

CERTIFICATIONS

- Microchip Technology Inc. Advance C programming
- EmbeddedExpertIO™ Advanced Embedded Systems Bare-Metal Programming Ground Up™
- UDEMY CMake, Tests and Tooling for C/C++ Projects [2022 Edition]
- edX Verified Certificate for AWS IoT: Developing and Deploying an Internet of Things